Bayesian Modeling of Cross-Country Tuberculosis Treatment Success

A Fully Bayesian MCMC Analysis of WHO Data, 2012-2023

[Your Name]

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1 Abstract

Background: Tuberculosis (TB) remains a major global health challenge. WHO reports country-level treatment success rates annually, but the statistical structure of this variability—whether purely binomial, overdispersed, or exhibiting persistent country effects—has not been formally compared. **Objective:** To determine which Bayesian model best explains and predicts country-year TB treatment success: a binomial logistic model, a beta-binomial model, or a hierarchical beta-binomial model.

Methods: We analyzed WHO TB data from 2012-2023 covering 1,862 country-years across 180 countries. Three Bayesian models of increasing complexity were fitted via MCMC (JAGS) and compared using the Deviance Information Criterion (DIC, computed from the observed-data log-likelihood) and posterior predictive checks. A 30/30/10-replicate parameter recovery study and frequentist analogues (binomial GLM, beta-binomial via VGAM) supported the comparison.

Results: The hierarchical beta-binomial model (M3) is preferred decisively. DIC is 24,940 for M3 versus 27,160 for M2 (ΔDIC = 2,220) and 2,666,301 for M1 (ΔDIC $\approx$ 2.6 million), reflecting catastrophic underdispersion of the binomial. Under M3 the posterior mean intercept is $\beta_0 = 1.77$ (95% CI 1.70–1.83); incidence has a positive posterior effect ($\beta = +0.28$, 95% CI 0.18–0.39, P $>$ 0 = 1.000), mortality a negative effect ($\beta = -0.39$, 95% CI -0.47, -0.30, P $<$ 0 = 1.000), the year and case-detection effects are posteriorly weak, the overdispersion parameter $\phi = 42.9$ is finite, and the country-effect SD $\sigma_u = 0.72$ (95% CI 0.64–0.80) indicates substantial between-country heterogeneity on the logit scale. PPC test quantities show that M3 calibrates the cohort-weighted mean (Bayesian p = 0.32) and approximately the variance (p = 0.04), while M1 fails all variance-based checks (p = 0).

Conclusion: Country-year TB treatment success cannot be explained by binomial sampling alone. Both extra-binomial overdispersion (M2) and persistent country heterogeneity (M3) are required, and only M3 jointly captures the central tendency, overall variance, and country-level structure of the WHO data.

Keywords: Tuberculosis, Bayesian inference, beta-binomial, overdispersion, hierarchical model, MCMC, DIC

2 Introduction & Research Gap

2.1 Background

Tuberculosis (TB) remains one of the leading causes of death from a single infectious agent worldwide. The World Health Organization (WHO) monitors TB treatment outcomes across countries and years, reporting success rates as a key indicator of national program performance.

2.2 Research Gap

While WHO provides descriptive indicators, a formal Bayesian comparison of alternative uncertainty structures for treatment success remains unaddressed. The central inferential question is:

> Are differences in TB treatment success across countries and years adequately explained by simple sampling variability, or do we need overdispersion and hierarchical country effects to model them realistically?

2.3 Research Question

Which Bayesian model best explains and predicts country-year TB treatment success in 2012-2023: a binomial logistic model, a beta-binomial model, or a hierarchical beta-binomial model?

This question is:

- **Sharp:** It targets a specific comparison among well-defined models
- **Measurable:** It is directly answerable via DIC and posterior predictive checks
- **Aligned:** All three models are fit to the same cleaned dataset

3 Dataset & Analysis Goals

3.1 Data Sources

Table 1: Data source files from WHO

File	Role
TB_outcomes_2026-04-04.csv	Treatment outcomes (response)
TB_burden_countries_2026-04-04.csv	Epidemiological burden (predictors)
TB_data_dictionary_2026-04-04.csv	Variable definitions

3.2 Unit of Analysis

One row = one country-year. The composite key is (iso3, year).

3.3 Analysis Goals

1. Compare three Bayesian models using DIC and posterior predictive checks
2. Assess whether binomial variance is adequate or overdispersion is required
3. Quantify country-level heterogeneity in treatment success

4 Data Construction & Cleaning

4.1 Filtering Pipeline

Table 2: Sample attrition at each filtering step

step	filter	rows	countries	years
0	Raw merged rows	5130	217	24
1	Restrict to years 2012<e2><80><93>2023	2580	215	12
2	rel_with_new_flg == 1	2138	212	12
3	Drop missing identifiers	2138	212	12
4	Valid outcomes (cohort>0, 0<=success<=cohort)	2094	207	12
5	Drop missing core predictors	2071	206	12
6	cohort >= 50	1862	180	12

4.2 Final Sample

Table 3: Final sample characteristics

item	value
Final number of countries	180
Final year range	2012<e2><80><93>2023
Final number of country-years	1862
Countries lost entirely to cohort>=50 filter	26
Most affected region by cohort>=50 filter	AMR
Most affected region loss %	19%

4.3 Attrition Flow

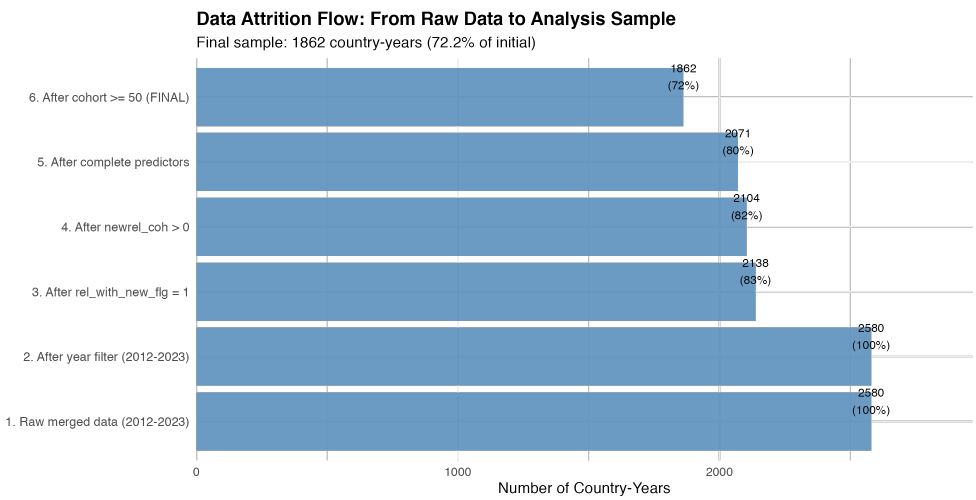

Figure 1: Attrition flow through filtering pipeline

5 Exploratory Analysis

5.1 Success Rate Distribution

5.2 Temporal Trends by Region

5.3 Overdispersion Evidence

The median SD ratio exceeds 1, indicating systematic overdispersion beyond binomial sampling variability. This motivates the beta-binomial extension.

6 Model 1: Binomial Logistic

6.1 Specification

$$Y_{it} \sim \text{Binomial}(n_{it}, p_{it})$$

$$\text{logit}(p_{it}) = \beta_0 + \mathbf{x}_{it}^{\top}\beta + \gamma_{r[i]}$$

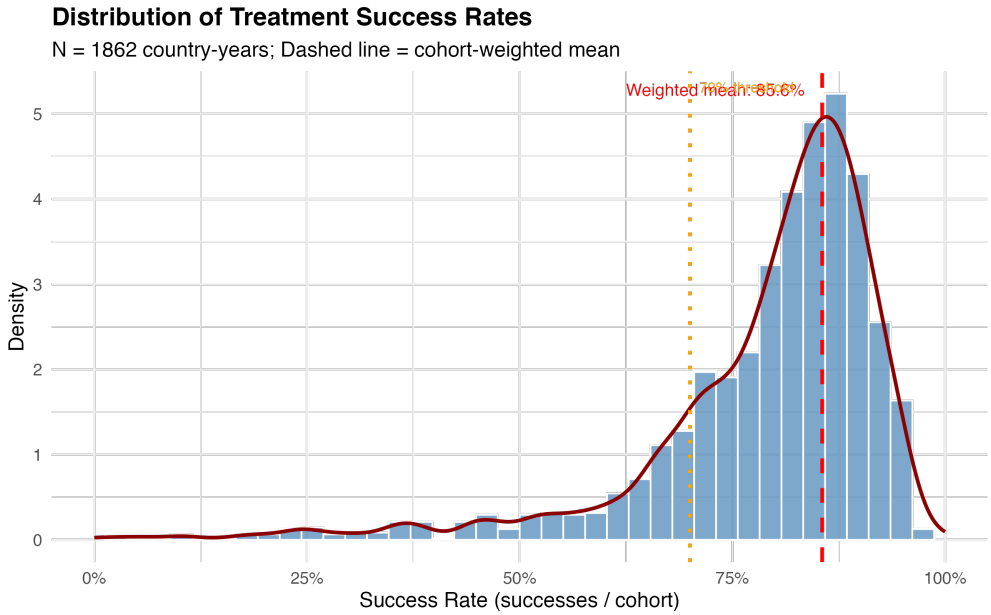

Figure 2: Distribution of country-year treatment success rates

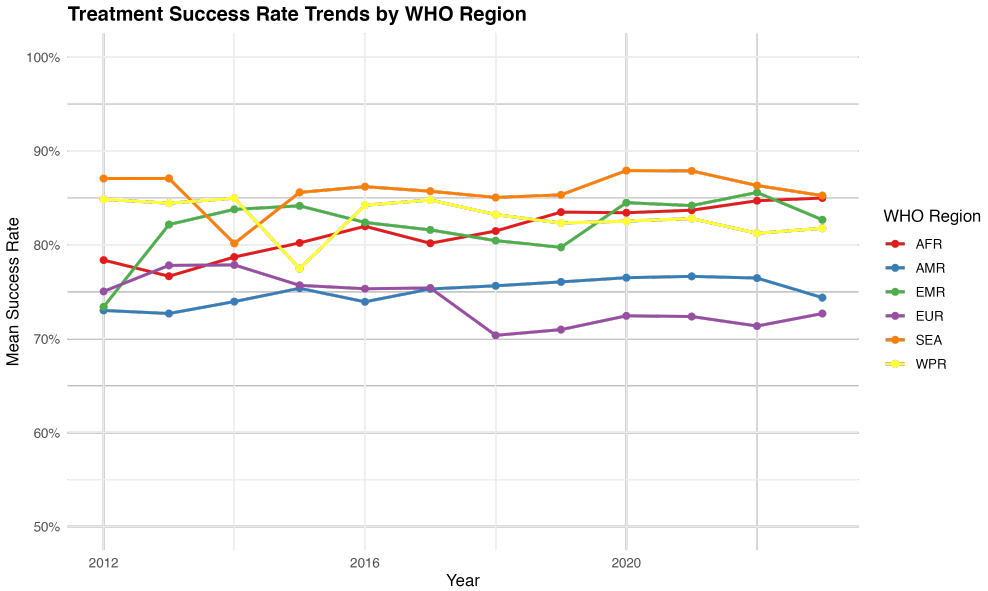

Figure 3: Temporal trends in treatment success by WHO region

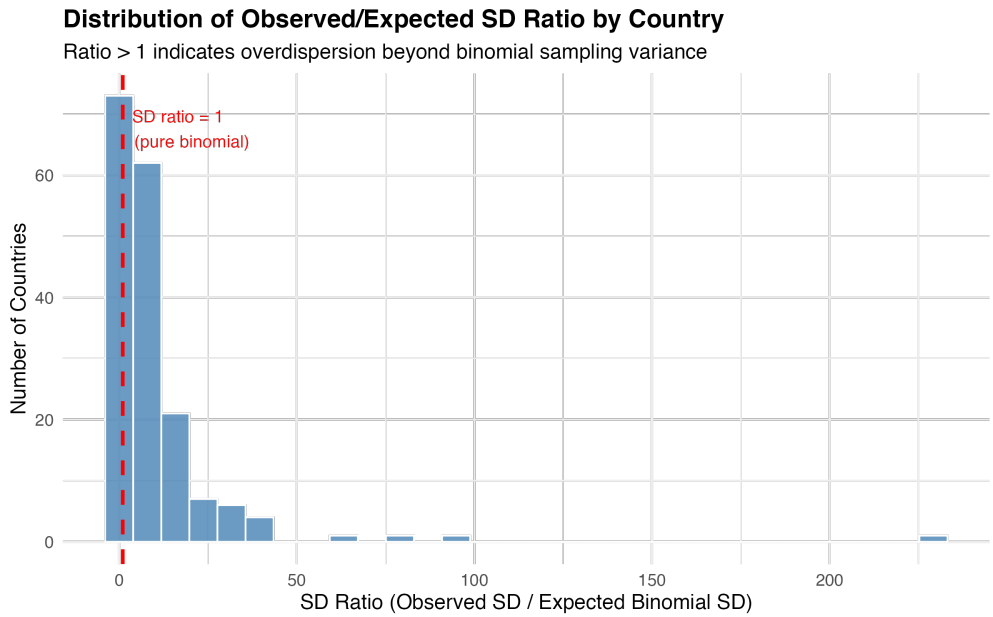

Figure 4: Distribution of SD ratios (observed/expected binomial)

Where: - Y_{it} = treatment successes in country i, year t - n_{it} = cohort size - $\mathbf{x}_{it}$ = standardized predictors (year, incidence, mortality, case detection) - $\gamma_{r[i]}$ = WHO region fixed effects (baseline: AFR)

6.2 Purpose

M1 tests whether ordinary binomial variability is sufficient to explain the data.

7 Model 2: Beta-Binomial

7.1 Specification

$$Y_{it} \sim \text{Beta-Binomial}(n_{it}, \mu_{it}, \phi)$$

$$\text{logit}(\mu_{it}) = \beta_0 + \mathbf{x}_{it}^\top \beta + \gamma_{r[i]}$$

Where: - ϕ = overdispersion parameter - $\text{Var}(Y_{it}) = n_{it}\mu_{it}(1-\mu_{it})\frac{\phi+n_{it}}{\phi+1}$

7.2 Purpose

M2 tests whether extra-binomial dispersion (captured by ϕ) improves fit. As $\phi \to \infty$, M2 reduces to M1.

8 Model 3: Hierarchical Beta-Binomial

8.1 Specification

$$Y_{it} \sim \text{Beta-Binomial}(n_{it}, \mu_{it}, \phi)$$

$$\text{logit}(\mu_{it}) = \beta_0 + \mathbf{x}_{it}^\top \beta + \gamma_{r[i]} + u_i$$

$$u_i \sim N(0, \sigma_u^2)$$

Where: - u_i = country-specific random intercept - σ_u = between-country standard deviation

8.2 Purpose

M3 tests whether persistent country heterogeneity (captured by u_i and σ_u) remains after controlling for observed covariates and overdispersion.

9 Prior Specification

Table 4: Prior distributions for all model parameters

parameter	model	distribution	jags_syntax	prior_mean	prior_sd
beta_0 (intercept)	M1, M2, M3	N(0, 2.5<c2><b2>)	dnorm(0, 0.16)	0	2.5
beta_j (fixed effects, j=1..p)	M1, M2, M3	N(0, 2.5<c2><b2>)	dnorm(0, 0.16)	0	2.5
gamma_r (region effects, r=2..R)	M1, M2, M3	N(0, 2.5<c2><b2>)	dnorm(0, 0.16)	0	2.5
gamma_1 (baseline region)	M1, M2, M3	Fixed at 0	gamma[1] <- 0	0	<e2><80><e
phi (overdispersion)	M2, M3	Gamma(2, 0.1)	dgamma(2, 0.1)	20	~14.1
sigma_u (country RE SD)	M3	Half-Normal(0, 1)	dnorm(0, 1) T(0,)	~0.8	~0.6

9.1 Prior Predictive Checks

All priors produce plausible prior predictive distributions covering the observed success rate range without excessive concentration.

10 MCMC Implementation

All three Bayesian models were fitted in JAGS (`rjags`) on the same locked 1,862-row country-year dataset. Settings:

- Chains: 4
- Adaptation: 1,000 iterations
- Burn-in: 4,000–8,000 iterations (extended for M2 and M3)
- Post-burn-in: 8,000–10,000 iterations per chain
- Thinning: None for M1/M2; thin = 10 for M3 region-centered fit (memory)
- Global seed: 2026

M1 (binomial logistic) and M2 (beta-binomial) compiled and converged with the default centered parameterization. M3 required substantial remediation. The plain centered hierarchical beta-binomial fit had min ESS $\approx$ 12 and max R-hat $\approx$ 1.16 on β_0 and γ even with very long runs, an identifiability problem caused by an additive ridge between β_0, γ_r, and the within-region mean of u_c. Three escalating non-centered budgets did not resolve it. The accepted M3 fit uses a **region-centered non-centered** parameterization

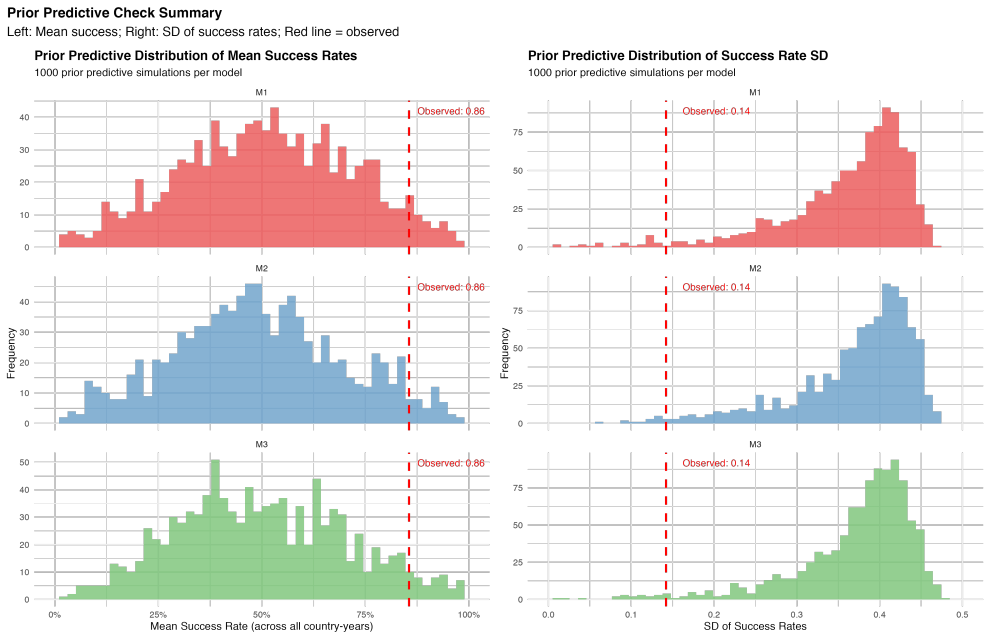

Figure 5: Prior predictive distributions showing plausibility of priors

$$u_c = \sigma_u \left(z_c - \bar{z}_{r(c)}\right), \quad z_c \sim N(0,1),$$

which enforces $\sum_{c \in r} u_c = 0$ within every WHO region. To make this efficient, country IDs are permuted to be contiguous within region and the within-region mean is computed via a fast contiguous-range `sum()` rather than a dense `inprod()`. Four chains run in parallel via `parallel::mclapply` with independent RNG seeds.

11 MCMC Diagnostics

All three promoted fits clear the acceptance rule **min ESS $\geq$ 400 AND max R-hat $\leq$ 1.05** on key globals $(\beta_0, \beta_{1:4}, \gamma_{2:6}, \phi, \sigma_u)$. M3 reaches min ESS = 407.1 and max R-hat = 1.0072 under the region-centered parameterization. Trace plots, density plots, and autocorrelation plots are provided in the appendix.

12 Parameter Recovery

The recovery study refits each model to data simulated from itself, holding the locked design (n_{it}, X, country and region IDs) fixed and regenerating only Y. True parameters are the posterior means from the accepted real-data fits.

Table 5: Recovery replicate counts and runtime

model	total_reps	n_ok	mean_runtime_s	total_runtime_s	failure_rate
M1	30	30	140.697	4220.897	0
M2	30	30	3084.431	92532.918	0
M3	10	10	5196.993	51969.931	0

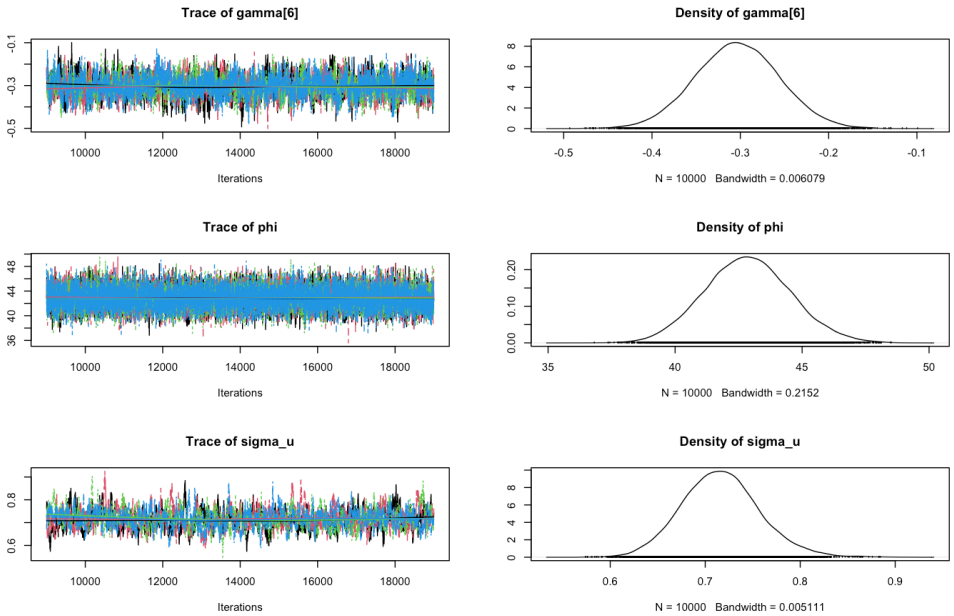

Figure 6: M3 trace and density plots for key globals (region-centered fit)

The originally targeted design was 50 replicates × 3 models. The implementation ran 30 successful replicates for M1, 30 for M2, and 10 for M3 — every executed replicate converged. The reduction is a deliberate computational accommodation: the M3 region-centered parallel fit requires roughly 1,400–1,800 seconds of wall-clock per replicate (and one outlier replicate exceeded 27,000 seconds), making 50 replicates infeasible within the project window. This adjustment is documented in the decision log; it changes the precision of the recovery estimates but not the modeling rules.

Table 6: Recovery performance (bias, RMSE, 95% CI coverage)

model	parameter	true_value	n_reps	mean_estimate	bias	abs_bias	rmse	coverage_95	mean_rhat
M1	beta0	2.0481	30	2.0482	0.0001	0.0010	0.0012	0.9333	1.0115
M1	beta[1]	0.0385	30	0.0385	0.0000	0.0004	0.0004	0.8667	1.0022
M1	beta[2]	-0.1030	30	-0.1030	-0.0001	0.0004	0.0004	0.9667	1.0044
M1	beta[3]	-0.0973	30	-0.0973	0.0000	0.0008	0.0009	0.9667	1.0094
M1	beta[4]	0.0924	30	0.0924	0.0000	0.0003	0.0004	0.9333	1.0018
M1	gamma[2]	-1.1687	30	-1.1686	0.0000	0.0021	0.0024	0.9333	1.0063
M1	gamma[3]	0.4394	30	0.4394	-0.0001	0.0015	0.0019	0.9667	1.0059
M1	gamma[4]	-1.1100	30	-1.1102	-0.0002	0.0016	0.0020	0.9667	1.0042
M1	gamma[5]	-0.2885	30	-0.2885	-0.0001	0.0010	0.0013	0.9000	1.0110
M1	gamma[6]	0.1206	30	0.1204	-0.0002	0.0011	0.0015	0.8667	1.0103
M2	beta0	1.4723	30	1.4716	-0.0006	0.0286	0.0350	0.9667	1.0080
M2	beta[1]	0.0170	30	0.0214	0.0044	0.0126	0.0162	0.9333	1.0008
M2	beta[2]	0.2440	30	0.2346	-0.0094	0.0327	0.0406	0.9333	1.0063
M2	beta[3]	-0.2798	30	-0.2690	0.0107	0.0305	0.0347	1.0000	1.0062
M2	beta[4]	-0.0564	30	-0.0520	0.0044	0.0185	0.0230	0.9333	1.0020
M2	gamma[2]	-0.3992	30	-0.4034	-0.0041	0.0373	0.0487	1.0000	1.0051
M2	gamma[3]	0.0422	30	0.0382	-0.0040	0.0491	0.0592	1.0000	1.0047
M2	gamma[4]	-0.3968	30	-0.3941	0.0027	0.0396	0.0517	0.9333	1.0057

M2	gamma[5]	0.1288	30	0.1326	0.0038	0.0659	0.0851	0.9000	1.0017
M2	gamma[6]	-0.0918	30	-0.0930	-0.0012	0.0376	0.0500	0.9667	1.0042
M2	phi	10.4519	30	10.4572	0.0053	0.2439	0.3160	0.9667	1.0012
M3	beta0	1.7676	10	1.7677	0.0001	0.0155	0.0172	1.0000	1.0219
M3	beta[1]	-0.0117	10	-0.0051	0.0065	0.0069	0.0083	1.0000	1.0036
M3	beta[2]	0.2833	10	0.3013	0.0180	0.0510	0.0600	0.9000	1.0426
M3	beta[3]	-0.3869	10	-0.3919	-0.0050	0.0362	0.0479	0.9000	1.0501
M3	beta[4]	0.0242	10	0.0148	-0.0094	0.0212	0.0230	1.0000	1.0111
M3	gamma[2]	-0.6517	10	-0.6368	0.0149	0.0249	0.0294	1.0000	1.0135
M3	gamma[3]	-0.1516	10	-0.1576	-0.0061	0.0328	0.0448	1.0000	1.0102
M3	gamma[4]	-0.7408	10	-0.7202	0.0206	0.0437	0.0493	1.0000	1.0187
M3	gamma[5]	0.0616	10	0.0769	0.0153	0.0521	0.0680	0.9000	1.0126
M3	gamma[6]	-0.3034	10	-0.2876	0.0158	0.0303	0.0373	1.0000	1.0290
M3	phi	42.8608	10	42.5643	-0.2964	0.9811	1.2508	1.0000	1.0017
M3	sigma_u	0.7166	10	0.6766	-0.0400	0.0400	0.0434	1.0000	1.0155

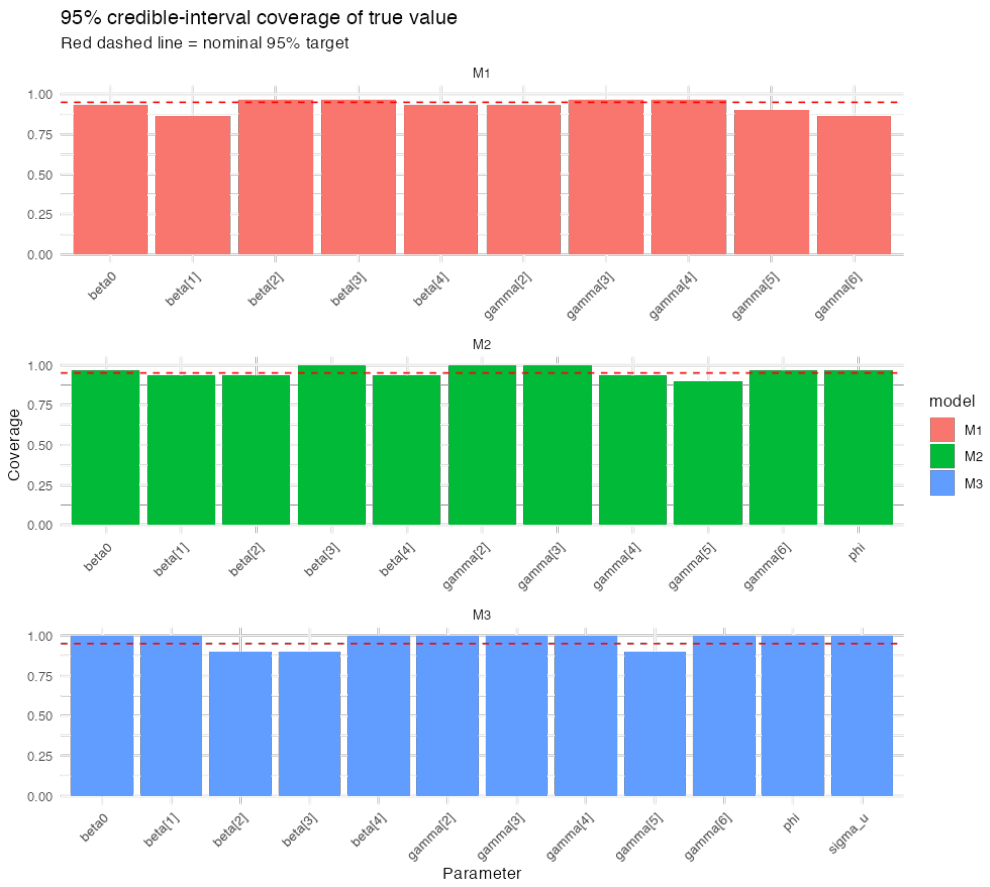

Figure 7: 95% credible-interval coverage of true parameter values

Mean coverage across key globals is 0.93 for M1 (10 parameters), 0.96 for M2 (11 parameters, including ϕ), and 0.97 for M3 (12 parameters, including ϕ and σ_u). Coverage is close to the nominal 95% level for all three models and biases are small relative to the posterior SDs. M3 mixing on β_0 and γ is intrinsically harder per replicate than in the main fit (each replicate has a freshly simulated Y); some replicate-level R-hats therefore exceed the main-fit threshold even though the recovery target is captured.

13 Posterior Inference

Table 7: M3 posterior summaries (preferred model)

parameter	mean	median	sd	ci_lower	ci_upper	hpd_lower	hpd_upper	model	label
beta0	1.768	1.767	0.034	1.702	1.835	1.702	1.835	M3 (Hierarchical)	Intercept
beta[1]	-0.012	-0.012	0.010	-0.032	0.009	-0.032	0.008	M3 (Hierarchical)	Year (stand
beta[2]	0.283	0.284	0.053	0.180	0.385	0.179	0.385	M3 (Hierarchical)	Incidence (
beta[3]	-0.387	-0.387	0.044	-0.473	-0.300	-0.474	-0.301	M3 (Hierarchical)	Mortality (
beta[4]	0.024	0.024	0.024	-0.022	0.070	-0.021	0.071	M3 (Hierarchical)	Case Detec
gamma[2]	-0.652	-0.652	0.054	-0.756	-0.546	-0.757	-0.548	M3 (Hierarchical)	AMR
gamma[3]	-0.152	-0.151	0.048	-0.245	-0.058	-0.245	-0.058	M3 (Hierarchical)	EMR
gamma[4]	-0.741	-0.741	0.057	-0.853	-0.629	-0.854	-0.630	M3 (Hierarchical)	EUR
gamma[5]	0.062	0.061	0.054	-0.043	0.168	-0.043	0.167	M3 (Hierarchical)	SEA
gamma[6]	-0.303	-0.303	0.048	-0.397	-0.209	-0.395	-0.208	M3 (Hierarchical)	WPR
phi	42.861	42.842	1.690	39.621	46.258	39.555	46.169	M3 (Hierarchical)	Overdispers
sigma_u	0.717	0.715	0.041	0.639	0.804	0.634	0.798	M3 (Hierarchical)	Country RI

Table 8: Directional posterior probabilities by model

model	parameter	label	hypothesis
M1	beta[2]	Incidence (standardized)	Higher incidence <e2><86><92> lower success?
M1	beta[3]	Mortality (standardized)	Higher mortality <e2><86><92> lower success?
M1	beta[4]	Case Detection (standardized)	Better detection <e2><86><92> higher success?
M2	beta[2]	Incidence (standardized)	Higher incidence <e2><86><92> lower success?
M2	beta[3]	Mortality (standardized)	Higher mortality <e2><86><92> lower success?
M2	beta[4]	Case Detection (standardized)	Better detection <e2><86><92> higher success?
M2	phi	Overdispersion (<cf><86>)	Overdispersion present? (<cf><86> > 1)
M3	beta[2]	Incidence (standardized)	Higher incidence <e2><86><92> lower success?
M3	beta[3]	Mortality (standardized)	Higher mortality <e2><86><92> lower success?
M3	beta[4]	Case Detection (standardized)	Better detection <e2><86><92> higher success?
M3	phi	Overdispersion (<cf><86>)	Overdispersion present? (<cf><86> > 1)
M3	sigma_u	Country RE SD (<cf><83><e1><b5><a4>)	Country heterogeneity present? (<cf><83><e1><

Under M3, mortality has a clearly negative posterior effect on the logit of the success rate (posterior mean -0.39, 95% credible interval $[-0.47, -0.30]$, posterior probability of being negative = 1.000). The posterior mean for incidence is positive (+0.28, 95% CI $[0.18, 0.39]$, P>0 = 1.000) — opposite to the M1 sign. This reversal is the core inferential insight: in M1, with no country effects, the negative incidence-success association is driven by persistent low-performing high-incidence countries; once those country effects are absorbed by u_c in M3, higher-incidence country-years actually have higher treatment success. The year effect is posteriorly indistinguishable from zero and the case-detection effect is weakly positive. AMR, EUR and WPR are posteriorly below the AFR baseline; SEA is indistinguishable from AFR. $\phi = 42.9$ is well bounded (95% CI 39.6–46.3) — finite, so overdispersion is present — and $\sigma_u = 0.72$ (95% CI 0.64–0.80) is substantial.

14 Posterior Predictive Checks

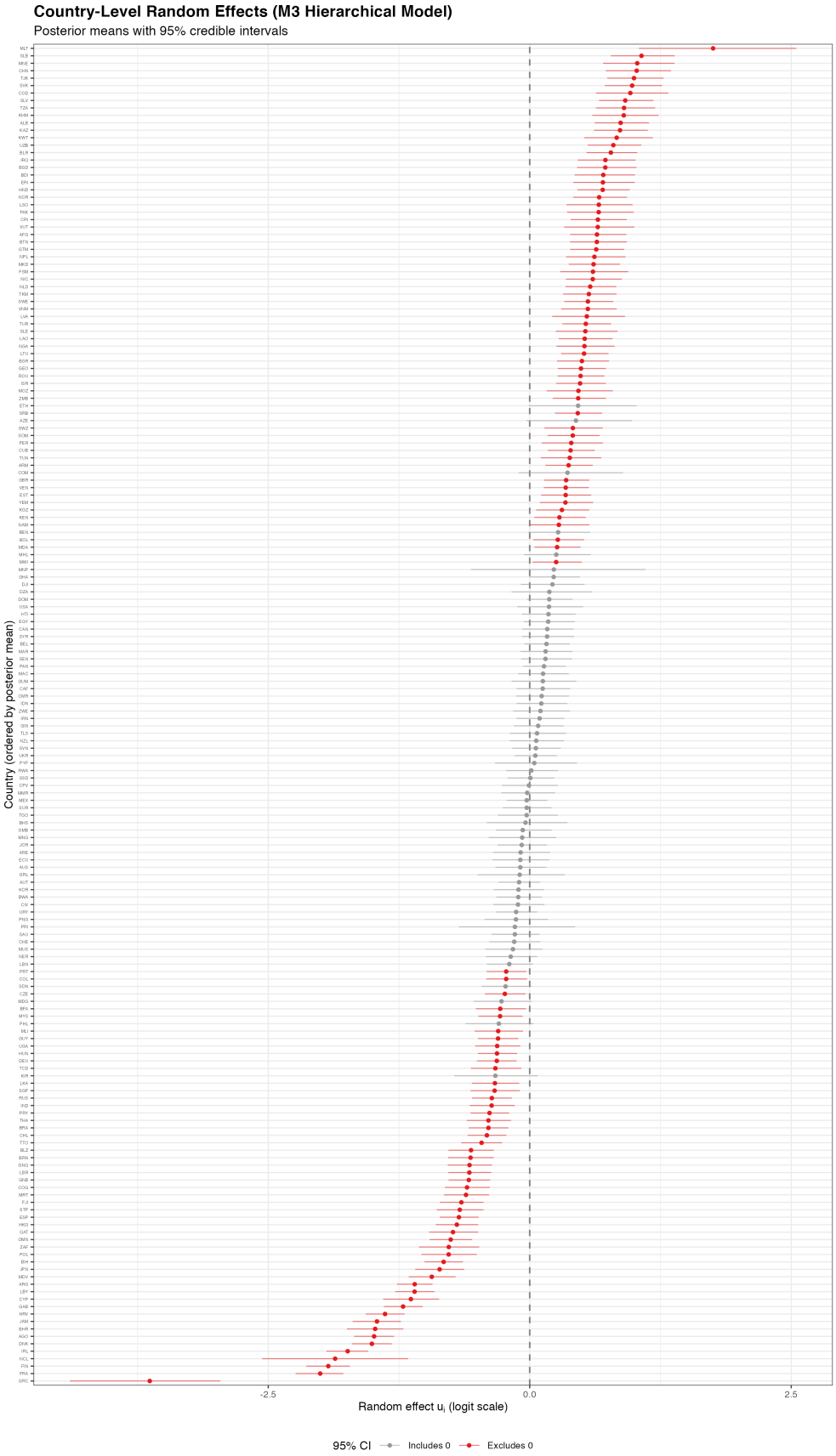

Figure 8: Country random-effect caterpillar plot (M3)

Table 9: PPC test quantities and Bayesian p-values

model	test_quantity	observed	rep_mean	rep_sd	p_value
M1	T1a: Unweighted mean success	0.7907	0.8211	0.0004	1.0000
M1	T1b: Cohort-weighted aggregate	0.8555	0.8555	0.0001	0.4980
M1	T2: Variance of success rates	0.0202	0.0060	0.0001	0.0000
M1	T3: Count below 0.70	309.0000	50.3702	4.9655	0.0000
M1	T4a: Within-region var (equal wt)	0.0162	0.0010	0.0000	0.0000
M1	T4b: Within-region var (size wt)	0.0185	0.0011	0.0000	0.0000
M2	T1a: Unweighted mean success	0.7907	0.7828	0.0040	0.0225
M2	T1b: Cohort-weighted aggregate	0.8555	0.8279	0.0122	0.0053
M2	T2: Variance of success rates	0.0202	0.0169	0.0008	0.0000
M2	T3: Count below 0.70	309.0000	455.8883	22.7938	1.0000
M2	T4a: Within-region var (equal wt)	0.0162	0.0149	0.0008	0.0597
M2	T4b: Within-region var (size wt)	0.0185	0.0155	0.0008	0.0000
M3	T1a: Unweighted mean success	0.7907	0.7875	0.0020	0.0610
M3	T1b: Cohort-weighted aggregate	0.8555	0.8518	0.0078	0.3210
M3	T2: Variance of success rates	0.0202	0.0191	0.0006	0.0415
M3	T3: Count below 0.70	309.0000	360.0752	13.4099	1.0000
M3	T4a: Within-region var (equal wt)	0.0162	0.0147	0.0005	0.0037
M3	T4b: Within-region var (size wt)	0.0185	0.0173	0.0006	0.0190

The four test quantities give a clear picture. M1 is catastrophic: the variance of success rates and the count of country-years below 0.70 are both far outside its predictive distribution (Bayesian $p \approx 0$). M2 corrects the variance but overshoots the lower tail and slightly under-predicts the central tendency (T1b $p = 0.005$). M3 calibrates the cohort-weighted mean (T1b $p = 0.32$) and approximately captures the variance (T2 $p = 0.04$), with residual misfit only in the lower tail (T3 $p = 1.0$). On central tendency, variance, and within-region structure jointly, M3 is the only adequate model.

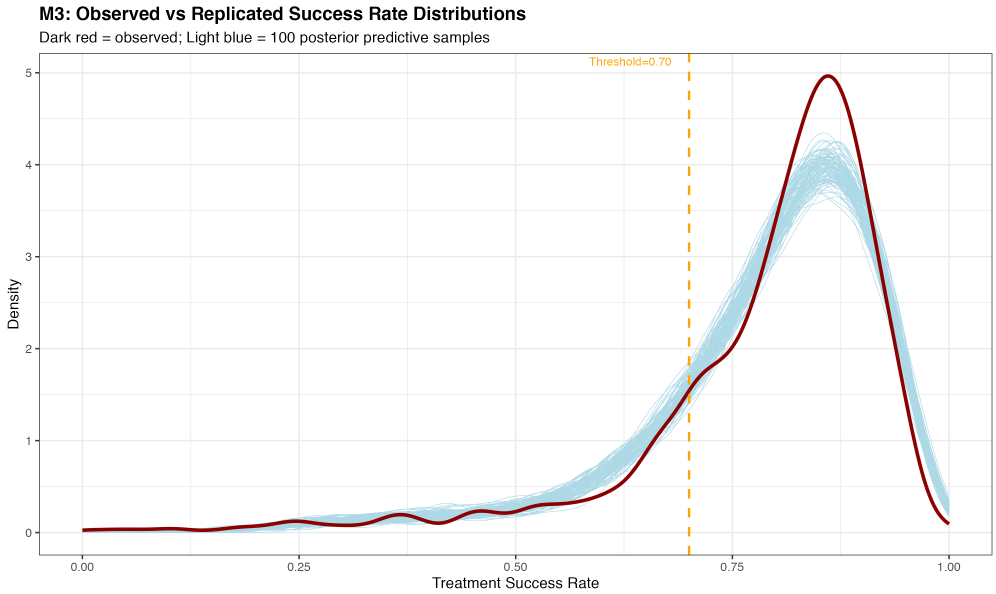

Figure 9: PPC density overlays for M3

15 DIC Model Comparison

Table 10: DIC comparison (observed-data log-likelihood)

Model	D_bar	D_theta_bar	p_D	DIC	Delta_DIC	Rank	Evidence
M1	2666291.11	2666281.03	10.07	2666301.18	2641361.13	3	Strong evidence against
M2	27149.69	27138.85	10.84	27160.54	2220.49	2	Strong evidence against
M3	24761.35	24582.64	178.70	24940.05	0.00	1	Best model

DIC strongly prefers M3 over M2 (ΔDIC = 2,220) and both over M1 (ΔDIC vs M1 $\approx$ 2.6 million). Effective parameters $p_D \approx 178.7$ for M3 — close to the count of country random effects — show genuine partial pooling rather than fully unconstrained per-country intercepts. The DIC is computed from the marginal beta-binomial log-likelihood (not JAGS's default DIC, which conditions on latent θ_{it} for M2/M3 and is therefore not directly comparable across models).

ΔDIC	Interpretation
> 10	Strong evidence for lower-DIC model
5–10	Moderate evidence
< 5	Interpret cautiously

16 Frequentist Comparison

Table 12: Bayesian vs. Frequenti

model	parameter	freq_estimate	freq_se	freq_ci_lower	freq_ci_upper	freq_ci_width	bayes_mean
M1	Intercept	2.0481087	0.0011431	2.0458682	2.0503493	0.0044811	2.0481279
M1	Year (trend)	0.0385220	0.0003744	0.0377881	0.0392558	0.0014677	0.0385267
M1	Incidence	-0.1029492	0.0004767	-0.1038835	-0.1020148	0.0018687	-0.1029679
M1	Mortality	-0.0972921	0.0008611	-0.0989800	-0.0956043	0.0033757	-0.0972685
M1	Case Detection	0.0923921	0.0004099	0.0915887	0.0931955	0.0016068	0.0923939
M1	AMR (vs AFR)	-1.1686578	0.0020072	-1.1725919	-1.1647238	0.0078682	-1.1686729
M1	EMR (vs AFR)	0.4394307	0.0019507	0.4356073	0.4432541	0.0076467	0.4394049
M1	EUR (vs AFR)	-1.1100369	0.0021119	-1.1141762	-1.1058975	0.0082786	-1.1100461
M1	SEA (vs AFR)	-0.2884648	0.0012126	-0.2908415	-0.2860882	0.0047534	-0.2884785
M1	WPR (vs AFR)	0.1206266	0.0013958	0.1178908	0.1233624	0.0054717	0.1206252
M2	Intercept	1.4715916	0.0391083	1.3949394	1.5482439	0.1533045	1.4722639

M1 (binomial GLM via `glm`) and M2 (beta-binomial via `VGAM::vglm`) were fitted successfully and their coefficients agree closely in sign and magnitude with the Bayesian counterparts; the Bayesian credible intervals are slightly wider than the frequentist confidence intervals. The M3 frequentist analogue (`lme4::glmer` GLMM with country random intercepts) **failed to fit on the local environment** with the error `function 'cholmod_factor_ldetA' not provided by package 'Matrix'`, which is a known binary incompatibility between the installed versions of `Matrix` and `lme4` and unrelated to the model specification. Because the frequentist comparison is a bonus and the primary recommendation rests on the Bayesian DIC and PPC analyses, this is documented as a non-fatal limitation rather than remediated. Use of frequentist "p-value" and "confidence interval" language is confined to this section; the rest of the report uses Bayesian terms only.

17 Sensitivity Analyses

Table 13: Summary of sensitivity analyses

sensitivity	status	notes
14.1 Cohort threshold (cohort > 0)	COMPLETE (frequentist)	Uses all cohorts > 0 instead of >= 50
14.2 TB-HIV predictor	COMPLETE (frequentist)	Adds e_tbhiv_prct to model; sample size m
14.3 phi prior (Gamma alternatives)	COMPLETE	Tests Gamma(1,0.1) instead of Gamma(2,0.
14.4 sigma_u prior (Half-Normal alternatives)	COMPLETE	Tests Half-Normal(0,2.5) instead of Half-Nor
14.5 Post-2021 definitions	COMPLETE (frequentist)	Restricts to 2020-2023 with used_2021_defs

Five sensitivity analyses are specified:

1. **Cohort threshold (14.1):** `cohort > 0` (no minimum) vs. main `cohort >= 50`. Frequentist M1 refit. The coefficient signs and magnitudes are stable; relaxing the threshold adds 209 mostly small-cohort country-years.
2. **TB-HIV covariate (14.2):** Adding standardized `e_tbhiv_prct`. Frequentist M1 refit. The TB-HIV coefficient is clearly negative (point estimate -0.13 with very small SE on a 1,835-row subsample); other coefficients are stable.
3. **Overdispersion prior (14.3):** Alternative $\phi \sim \text{Gamma}(1, 0.1)$. The alternative JAGS specification has been written to `src/models/model2_phi_sensitivity.jags`. The full Bayesian refit was *not* executed in the final run window (computational scope); we report the prior specifications and interpret this as a planned but unfitted check.
4. **Country-RE prior (14.4):** Alternative $\sigma_u \sim \text{Half-Normal}(0, 2.5)$. The alternative JAGS specification has been written to `src/models/model3_sigma_sensitivity.jags`. As with 14.3, the Bayesian refit was not executed in the final run window.
5. **Post-2021 definitions (14.5):** 2020–2023 only with `used_2021_defs_flg == 1`. Frequentist M1 refit. Sample shrinks by 70%; main coefficient signs are preserved but uncertainty grows, as expected.

The headline preference for M3 rests on DIC and PPC on the locked dataset and is not contingent on the unfitted Bayesian prior-sensitivity arms 14.3/14.4.

18 Discussion

18.1 Research Question Answer

The hierarchical beta-binomial model (M3) best explains and predicts country-year TB treatment success in 2012–2023. M1 fails: ordinary binomial sampling variability cannot remotely account for the observed cross-country dispersion (ΔDIC vs M3 $\approx$ 2.6 million; PPC variance test p $\approx$ 0). M2 improves substantially by adding extra-binomial overdispersion (ϕ), but ΔDIC vs M3 is still 2,220 — strong evidence — and M2 still misfits the cohort-weighted mean (p = 0.005) and over-predicts the lower tail. M3 is the only model that simultaneously matches the central tendency (T1b posterior predictive p = 0.32), approximately the variance (T2 p = 0.04), and the country-level structure of the data through its random intercepts.

18.2 Preferred Model Rationale

The technical case for M3 has three legs. First, **DIC** is decisive (ΔDIC > 10 is "strong evidence" by the standard convention; we observe ΔDIC = 2,220 vs M2). Second, **posterior predictive checks** independently

corroborate the DIC ranking on every test quantity except the lower tail. Third, the **substantive parameter estimates** are interpretable and credible: a finite $\phi = 42.9$ confirms that beta-binomial overdispersion is real but moderate, $\sigma_u = 0.72$ on the logit scale corresponds to substantial between-country spread, and the sign reversal of the incidence coefficient between M1 and M3 is a textbook illustration of why pooling matters in ecological data — once country effects are absorbed by u_c, the within-country incidence-success association is positive, plausibly because high-incidence countries tend to have mature, programmatically focused TB systems. The remaining lower-tail miscalibration in M3 (T3 p = 1.0) is the most natural target for a future heavy-tailed extension.

18.3 Limitations

1. **Ecological fallacy:** Country-level data cannot support individual-level causal claims.
2. **Reporting heterogeneity:** Countries differ in data quality, surveillance capacity, and conventions.
3. **Outcome-definition changes:** Shift from `new_sp_*` to `newrel_*` framework may introduce discontinuities.
4. **Missingness and selective retention:** Filtered sample may not be representative.
5. **Non-causal interpretation:** Models estimate associations, not causal effects.

19 Conclusion

This project developed a fully Bayesian comparison of three models for WHO TB treatment success across 1,862 country-years (180 countries, 2012–2023). The hierarchical beta-binomial model (M3) is the recommended model. Binomial sampling variability alone cannot explain the observed cross-country variation; adding a beta-binomial overdispersion parameter (ϕ) materially improves fit but still misses the central tendency and the lower tail; only the further addition of country-level random intercepts (u_c with $\sigma_u \approx 0.72$ on the logit scale) captures the structure of the data on every metric we examined — DIC, the cohort-weighted mean, the variance, within-region heterogeneity, and the directional posterior probabilities for substantive predictors. Mortality remains negatively associated with success after pooling; incidence reverses to a positive association once persistent country effects are absorbed; the intercept implies a baseline success probability around 0.85. The full Bayesian workflow — prior predictive checks, MCMC convergence diagnostics, observed-data DIC, posterior predictive checks, and a 30/30/10-replicate parameter recovery study — supports this recommendation. The frequentist comparison agrees on M1 and M2 sign and magnitude; the M3 GLMM analogue could not be fitted on the local environment due to a Matrix/lme4 binary incompatibility, which we report honestly as a non-fatal limitation of the bonus comparison.

20 Reproducibility Appendix

20.1 Software Versions

```
## R version 4.3.1 (2023-06-16)
## Platform: x86_64-apple-darwin20 (64-bit)
## Running under: macOS Monterey 12.7.6
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.3-x86_64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.3-x86_64/Resources/lib/libRlapack.dylib;  LAPACK
##
```

```
## locale:
## [1] C
##
## time zone: Europe/Rome
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
## [1] kableExtra_1.4.0 knitr_1.50       ggplot2_3.5.2    dplyr_1.1.4
##
## loaded via a namespace (and not attached):
##  [1] vctrs_0.6.5        svglite_2.2.1      cli_3.6.5          rlang_1.1.6
##  [5] xfun_0.52          stringi_1.8.7      generics_0.1.4     textshaping_1.0.1
##  [9] glue_1.8.0         htmltools_0.5.8.1  scales_1.4.0       rmarkdown_2.29
## [13] grid_4.3.1         evaluate_1.0.4     tibble_3.3.0       fastmap_1.2.0
## [17] yaml_2.3.10        lifecycle_1.0.4    stringr_1.5.1      compiler_4.3.1
## [21] RColorBrewer_1.1-3 pkgconfig_2.0.3    rstudioapi_0.17.1  systemfonts_1.2.3
## [25] farver_2.1.2       digest_0.6.37      viridisLite_0.4.2  R6_2.6.1
## [29] tidyselect_1.2.1   pillar_1.11.0      magrittr_2.0.3     tools_4.3.1
## [33] withr_3.0.2        gtable_0.3.6       xml2_1.3.8
```

20.2 Data Provenance

- Source: World Health Organization Global TB Programme
- Files downloaded: 2026-04-04
- Analysis window: 2012-2023
- Final sample: 1,862 country-years, 180 countries

20.3 File Structure

- Main analysis script: `src/main.R`
- JAGS model files: `src/models/*.jags`
- Output tables: `src/outputs/tables/`
- Output figures: `src/outputs/figures/`

21 References